



Wnode (Wenode) Technical Position Summary

Wenode has already achieved the hardest part: a functioning, end-to-end MVP that proves the architecture works in the real world. The orchestration engine, node operator app, WASM execution path, routing logic, and telemetry pipeline are all live, tested, and operating as a coherent system.

The next steps are not about building the network from scratch, they are about evolving, strengthening, and scaling what already works. Every item in the technical TODO is an enhancement, refinement, or extension of existing capabilities.

This is a mature foundation, not an early prototype. The system is ready to grow as soon as additional development capacity and hosting infrastructure are available.

Wenode is past the “can this be built?” phase. We are now in the “scale, harden, and optimize” phase where real networks become real businesses.

1. Orchestration Engine

- Evolve deterministic job routing — improve reproducibility, reduce jitter, tighten WASM execution paths.
- Enhance trust-scoring v2 — expand behavioral signals, add anomaly detection, refine scoring weights.
- Improve latency-aware routing — micro-region clustering, jitter smoothing, fallback heuristics.
- Upgrade predictive workload placement — integrate historical patterns with real-time telemetry.
- Strengthen autonomous optimization loop — improve feedback cycles, reduce oscillation, add self-tuning parameters.

2. Zero-Trust Compute Layer

- Upgrade WASM sandbox — stricter capability boundaries, improved syscall filtering.
- Extend micro-VM isolation — fallback to Firecracker-class isolation for high-risk workloads.
- Refine zero-storage execution — enforce RAM-only workloads with deterministic cleanup.

- Improve state validation — pre/post execution checks, signature verification, deterministic hashing.

3. Execution Runtime

- Evolve workload container format — WASM + metadata + resource caps.
- Improve resource limit enforcement — CPU, RAM, execution time, thermal constraints.
- Strengthen signature verification — multi-layer verification + tamper detection.
- Enhance crash recovery — auto-restart, state restore, failure telemetry.

4. Telemetry & Metrics

- Extend node heartbeat protocol — richer signals, reduced false positives.
- Improve execution metrics — latency, success/failure, resource usage, anomaly flags.
- Refine node performance scoring — multi-factor scoring with dynamic weighting.
- Enhance routing feedback signals — real-time routing hints + congestion indicators.

5. Networking & Routing

- Improve region-aware routing — better geo clustering + fallback logic.
- Strengthen fallback routing — multi-path routing + degraded mode.
- Add congestion detection improvements — detect hotspots, reroute proactively.
- Refine routing table persistence — faster rebuilds, better caching.

6. Node Operator App (Core)

- Enhance secure device registration — stronger binding between device → WUID → session.
- Improve encrypted config store — zero-plaintext, key-derived encryption.
- Upgrade workload execution module — better WASM runner, tighter resource caps.
- Extend telemetry sender — richer metrics, lower overhead.
- Improve update mechanism — delta updates, signature verification, rollback support.
- Strengthen crash recovery — auto-restart, state restore, local diagnostics.
- Refine headless mode — stable CLI flags, auto-start compute.
- Improve tray mode — background operation with minimal UI.

- Enhance capability detection — CPU, RAM, GPU, thermal limits.
- Improve session refresh — silent token renewal, failure fallback.
- Strengthen workload sandboxing — WASM + micro-VM fallback.
- Improve local logs — encrypted, rotating, minimal retention.
- Refine uninstall cleanup — remove keys, configs, cache

7. Developer SDK (Core)

- Improve job submission API — better validation + error handling.
- Enhance job status polling — reduce load, improve accuracy.
- Refine result retrieval — faster, more reliable, better error codes.
- Expand error code documentation — clearer developer guidance.